

Operating System - Questions and Answers

Q1 — Definition: What is an operating system (OS)? List four principal functions of an OS.

An Operating System (OS) is software that acts as an interface between the user and computer hardware. Functions: (1) Process management, (2) Memory management, (3) File management, (4) Device (I/O) management.

Q2 — Types: Give three classifications of operating systems and one example for each.

1. Based on user interface: GUI (Windows 11), Command-line (MS-DOS). 2. Based on task management: Single-tasking (MS-DOS), Multi-tasking (Linux). 3. Based on user access: Single-user (Windows), Multi-user (Unix).

Q3 — Process lifecycle: Describe the process state diagram and explain transitions New → Ready → Running → Waiting → Terminated.

States: New → Ready → Running → Waiting → Terminated. Transitions: New→Ready (admitted), Ready→Running (CPU scheduler dispatches), Running→Waiting (I/O request), Waiting→Ready (I/O completed), Running→Terminated (finished).

Q4 — CPU scheduling concept: What is turnaround time? How is it measured?

Turnaround time is the total time taken from submission to completion. Formula: Turnaround Time = Completion Time – Arrival Time.

Q5 — FCFS scheduling calculation example.

Processes: P1=120, P2=80, P3=100, P4=30, P5=160 (arrival=0). Completion Times: 120, 200, 300, 330, 490. Turnaround Times: 120, 200, 300, 330, 490. Waiting Times: 0, 120, 200, 300, 330. Average Turnaround = 288 ms, Average Waiting = 190 ms.

Q6 — Explain preemptive vs non-preemptive scheduling. Give one algorithm for each.

Preemptive: CPU can be taken from a process (e.g., Round Robin). Non-preemptive: CPU not taken until completion or I/O (e.g., FCFS).

Q7 — Logical vs Physical addresses: If frame size = 4KB, PMT maps page 2→frame 11, offset=1536, find the physical address.

Physical Address = (Frame × Frame Size) + Offset = (11 × 4096) + 1536 = 46080 bytes.

Q8 — Fixed vs variable partitioning: Explain one advantage and disadvantage each.

Fixed-size: Simple, but internal fragmentation. Variable-size: Efficient use of memory, but external fragmentation.

Q9 — Memory allocation methods: Describe best-fit, first-fit and worst-fit methods briefly.

Best-fit: Allocates smallest hole large enough (creates small holes). First-fit: Allocates first available hole (fast). Worst-fit: Allocates largest hole (leaves big remainders).

Q10 — Coalescing: Explain what coalescing of memory holes means and why it helps.

Coalescing combines adjacent free memory holes into one larger block to reduce external fragmentation.

Q11 — Paging concepts: Define page, frame, and page-map table (PMT).

Page: Fixed-size logical block. Frame: Fixed-size physical block. PMT: Maps pages to frames.
Physical Address = (Frame number × Frame size) + Offset.

Q12 — Process control: A process in Waiting state is waiting for what? Why is it not in Ready?

Waiting for I/O or event (e.g., resource). Not in Ready because it cannot continue until the event occurs.